



Vu Hai Duong

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📌 Duong Hai Vu

TECHNICAL SKILLS

Languages: Python, R

Tools: Git, GitHub, Docker, SQL Server, Anaconda

Frameworks & Libraries: PyTorch, TensorFlow, Pandas, NumPy, OpenCV, Langchain, FastAPI

Cloud/Databases: IBM Cloud

Spoken Languages: Vietnamese (Native), English (B2)

EXPERIENCE

• FPT University AI Research Laboratory

Mar 2024 – Feb 2025

Research Trainee

- Optimized EEG data preprocessing pipelines using advanced signal filtering techniques, improving feature extraction and reducing noise, leading to more robust model performance.
- Collaborated with interdisciplinary teams to refine model architectures and validate results using neuroscience benchmarks.
- Supervised by: **Professor Phan Duy Hung**.

• C.P. Vietnam Corporation

Apr 2025 – Sep 2025

Intern

- Worked on-site at the C.P. Vietnam Corporation – Food Processing Plant in Hanoi.
- Performed data labeling and annotation for computer vision tasks.
- Trained object detection models using YOLOv8 for pig detection in the Counter_Pig project.
- Developed an OCR pipeline for Vietnamese receipts using PaddleOCR for text detection and VietOCR for Vietnamese text recognition.

PROJECTS

• Deepfake Detection 🔗

Mini Team Project

- Tools & technologies used: **Python, TensorFlow, OpenCV, Deep Learning**
- Developed a **deepfake detection pipeline** to classify videos as real or manipulated using deep learning models.
- Implemented **video preprocessing, face extraction, and model training** to improve detection reliability on facial forgery datasets.

• Film Recommendation System 🔗

Personal Project

- Tools & technologies used: **Python, Pandas, Scikit-learn, NLTK, VADER, TF-IDF, MLxtend**
- Built a **hybrid recommendation system** combining **content-based filtering** and **collaborative filtering** to generate personalized movie suggestions.
- Performed **EDA, text mining, and sentiment analysis** on movie overviews and user reviews using TF-IDF vectorization and cosine similarity.

• AgriRAG: A Lightweight Vision-Language Framework for Rice Leaf Disease Recognition Using RAG 🔗

Mini Team Project | Python, EfficientNetB0, MobileNetV3, SqueezeNet, FAISS, BM25, RRF, LLaMA 3.3, FastAPI

- Developed an end-to-end **rice leaf disease recognition system** classifying **10 disease classes** across 15,023 images; achieved **99.01% test accuracy** (macro F1 = 0.989) using EfficientNetB0 with two-stage transfer learning.
- Applied **Focal Loss** ($\gamma=2.0$) and inverse-frequency class weighting to handle a 1.8:1 class imbalance, improving minimum per-class F1 by **+0.046** over standard cross-entropy baseline.
- Deployed FP16-quantised TFLite model achieving **74 ms inference latency** and **~195 MB RAM** on a mid-range Android device (Snapdragon 680), enabling real-world mobile deployment.
- Designed a **Hybrid RAG pipeline** combining FAISS dense retrieval, BM25 keyword search, and Reciprocal Rank Fusion with Cross-Encoder re-ranking; evaluated via RAGAS framework achieving **faithfulness = 0.91** and **answer relevance = 0.88** on a 100-question expert-verified benchmark.
- Implemented **Smart Chunking** (structure-aware segmentation by symptom/cause/treatment) and Parent-Child Retrieval over a **~340-page** agricultural corpus; served via **FastAPI + SSE** streaming backend with LLaMA-3.3-70B for grounded, citation-backed treatment recommendations.

EDUCATION

- **FPT University** 2022 - 2026
Bachelor's Degree of Artificial Intelligence CGPA: 3.42/4.0
- **Xuan Mai High School** 2019 - 2022
Student GPA: 9.2/10

COURSES AND CERTIFICATION

- **Machine Learning Specialization, Deep Learning Specialization, IBM Full Stack Software Developer Specialization**
- **Natural Language Processing, , Applied Data Science with R Specialization**

ACHIEVEMENTS

- **Honorable Student** of Trimester Fall 2024, Spring 2025, Summer 2025, Spring 2026
- **Excellent Student** of Trimester Fall 2025